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Warangal, Telangana, India

PROJECT REPORT

Classifying imagined shapes (circle/square/pentagram) from EEG signals

Stability-aware selection of channels, frequency bands and time windows
Final Report

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Abstract

Electroencephalography (EEG)-based visual-imagery brain–computer interfaces try to infer the shape a person is imagining from their brain activity. We tested whether a *stability-aware multi-objective genetic algorithm* (NSGA-II) that jointly selects EEG channels, frequency bands and time windows for a filter-bank common-spatial-pattern (FBCSP) and shrinkage-LDA classifier decodes imagined shapes better than the same classifier on all components. We used the public visual-imagery dataset of Gao et al. (22 participants, 42 sessions; circle, square and pentagram; chance = $1/3$) under a strict protocol: leave-one-block-out and cross-session evaluation, with every selection nested inside the training data and a hypothesis pre-registered before the results were analysed. **The primary hypothesis is not supported:** the GA reached a balanced accuracy of 0.345 against 0.353 for the full baseline ($\Delta = -0.007$, one-sided Wilcoxon $p = 0.79$, $n = 22$). No pre-registered secondary comparison was significant: an accuracy-only GA, cross-session evaluation, and EEGNet with 32 or 8 channels all gave Holm-corrected $p = 1.00$. The GA matched the baseline's accuracy at about 30 % of the cost (13 of 168 features), but exploratory controls show that its selection carries no information. Random subsets of the same size perform equally well, selections are as unstable as random subsets (Kuncheva index 0.08), and every model fits its training data (training accuracy 0.58–1.00) yet generalises to about chance +0.02. On the visual *perception* period, where the signal is decodable (0.435), the GA gives up a significant 0.025 in accuracy. We conclude that the imagery period of this dataset carries too little transferable class information for component selection to help, and we report this negative result with full reproducibility.

Keywords: EEG, visual imagery, brain–computer interface, FBCSP, NSGA-II, feature selection, EEGNet, pre-registration, reproducibility.

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1 Introduction

1.1 Problem statement

A brain–computer interface (BCI) provides a communication pathway based on brain activity rather than overt movement. In *visual imagery*, a participant mentally pictures a shape and a classifier tries to infer the shape from the recorded EEG epoch. Imagined-shape signals are weak and vary across people, sessions, time intervals and frequency bands. The central challenge, as stated in our proposal, is *to identify reliable, discriminative information without selecting features that work only for one validation split*.

1.2 From the proposal to the experiment

Our literature-review proposal targeted four classes $\mathcal{Y} = \{\text{circle}, \text{square}, \text{triangle}, \text{hexagon}\}$. It also noted that the most suitable public benchmark, the 2026 visual-imagery dataset of Gao et al. [6], contains circle, square and *pentagram* but not triangle or hexagon. No compatible four-class dataset became available, so, as the proposal anticipated, the study was carried out on the clearly labelled **three-class benchmark** $\mathcal{Y} = \{\text{circle}, \text{square}, \text{pentagram}\}$ (chance = 33.3 %). The method, evaluation and software are independent of the class set and apply unchanged to four classes once such data exist.

1.3 Objectives

1. Formulate the joint selection of electrodes, frequency bands and time windows as a *stability-aware multi-objective* search (accuracy, repeatability, cost).
2. Compare it, under a strict and pre-registered protocol, with matched baselines: the same classifier on all components, an accuracy-only GA, and the deep network EEGNet.
3. Test whether the search yields a more compact *and* more repeatable representation, and explain the outcome with controls.

1.4 Contributions

- A complete, tested pipeline (data download, preprocessing, FBCSP, NSGA-II, EEGNet, analysis) with fixed module interfaces and 200 automated tests.
- A pre-registered evaluation on 22 subjects with nested, leakage-controlled splits; the results are bit-identical across three machines.
- A clear negative result for the proposed method, and exploratory controls that explain it: matched-cost random subsets, alternative chooser rules, training-vs-test accuracy, trial-position confounds, random-split inflation and the perception (viewing) period.

2 Related Work

Table 1 summarises the six studies reviewed in our proposal. Their accuracies come from different datasets, class sets and protocols and do not rank the methods.

GAs [3], spatial filters [2, 5] and temporal windows [5] have all been used before, so none of these elements is novel on its own. The defensible distinction of this project, formulated in the proposal, is to treat *electrode, band and window selection as one optimisation problem* that weighs accuracy, repeatability and feature economy, and to test it experimentally against matched baselines. Our building blocks are FBCSP [7], NSGA-II [8] as implemented in pymoo [9], shrinkage LDA [10], and EEGNet [11] as implemented in braindecode [12].

Table 1: Selected visual-imagery EEG studies (from the literature review).

Study	Method and dataset	Main finding / relevance
Esfahani & Sundararajan (2012) [1]	ICA, Hilbert–Huang spectra, LDA; five imagined 3D primitives	44.6 % mean (chance 20 %); spectral-feature baseline
Costa et al. (2019) [2]	Alpha-band CSP, variance features, SVM; seven 2D figures	52 % mean; includes circle, square, triangle, hexagon
Llorella et al. (2020) [3]	CNN with GA parameter search; four objects + rest	60 %; prior use of GAs in visual imagery
Llorella et al. (2021) [4]	CNN with Black-Hole search; seven geometric shapes	35 % mean; multi-class difficulty
Korik et al. (2024) [5]	FBCSP, sliding windows, mutual information, regularised LDA; five 3D primitives	Closest pipeline; feedback improved multi-session performance
Gao et al. (2026) [6]	Public 32-channel, 1000 Hz dataset, 22 participants	EEGNet 75.1 % on three figure classes (within-subject)

3 Dataset and Preprocessing

3.1 Dataset

We used the figure task of the Gao et al. visual-imagery dataset [6] (figshare DOI 10.6084/m9.figshare. 30227503):

- 22 participants with up to two sessions; sub-09 and sub-10 have only session 1, giving **42 recordings** (20 subjects for cross-session analysis);
- 32 EEG channels at 1000 Hz (reference CPz); 120 trials per session (40 per shape) in 4 blocks of 30;
- trial timeline: fixation 0–3 s, **viewing** 3–7 s, mask 7–9 s, **imagery** 9–13 s, rest 13–17 s. The only event marker is the imagery onset.

3.2 Preprocessing

1. Notch filter at 50 Hz and its harmonics; 1–40 Hz zero-phase band-pass; common average reference; resampling to 250 Hz.
2. Imagery epochs of 4 s (1000 samples) from the marker; viewing epochs from 3–7 s for a sanity check.
3. Artefact rejection: a trial is dropped if its peak-to-peak amplitude on the 8 candidate channels exceeds 150 μV. This removed **126 of 5040 trials (2.5 %)**; the smallest class in any recording kept 33 trials.
4. Three caches per recording: band-filtered windows for FBCSP, a broadband epoch for EEGNet, and the viewing period. A reference manifest checks that every machine built identical caches.

Trial-order structure. Within each block the 30 trials (10 per shape) follow a fixed pseudo-random order. Only 19 distinct 120-trial sequences are used across the 42 recordings. Subject 01 repeats its whole sequence across sessions, and we found that 30-trial block orders recur across sessions for 10 of 20 subjects (Section 7.4). This motivates the strict block-wise evaluation.

4 Methodology

4.1 Component space and chromosome

The signal is decomposed into components that can be switched on or off:

- **8 channels** over the parieto-occipital (visual) cortex: O1, Oz, O2, Pz, P3, P4, P7, P8;
- **4 bands**: theta (4–7 Hz), alpha (8–12 Hz), beta (13–30 Hz), low gamma (30–40 Hz);
- **7 windows** of 1 s starting at 0, 0.5, …, 3.0 s after imagery onset.

A candidate solution is a 19-bit chromosome $b = (b^{ch}_{1..8}, b^{band}_{1..4}, b^{win}_{1..7})$. It is valid if at least 2 channels, 1 band and 1 window are selected; invalid offspring are repaired by switching on random genes. Its **cost** is the mean fraction of each group used:

$$c(b) = \frac{1}{3}\left(\frac{n_{ch}}{8} + \frac{n_{band}}{4} + \frac{n_{win}}{7}\right) \in (0, 1]. \tag{1}$$

4.2 Classifier: FBCSP with shrinkage LDA

For every selected band–window pair, one-vs-rest common spatial patterns [7] find spatial filters w that maximise the variance ratio between one class and the rest,

$$w^* = \arg\max_w \frac{w^\top \Sigma_c\, w}{w^\top (\Sigma_c + \Sigma_{\bar{c}})\, w}, \tag{2}$$

where Σ are class covariance matrices of the selected channels. One filter pair per class gives 6 log-variance features per band–window, so the full chromosome yields $4 \times 7 \times 6 = 168$ features. A linear discriminant with Ledoit–Wolf shrinkage [10] (`lsqr` solver) classifies the features. It stays well-conditioned with about 90 training trials.

4.3 Proposed method: stability-aware NSGA-II (GA-NSGA)

NSGA-II [8, 9] evolves a population of chromosomes towards the Pareto front of three objectives, each computed with a 3-fold inner cross-validation (CV) that uses *only the outer training blocks*:

$$\max\ \mu_{in}(b), \qquad \min\ \sigma_{in}(b), \qquad \min\ c(b), \tag{3}$$

where μ_{in} and σ_{in} are the mean and standard deviation of the inner-fold balanced accuracy. Minimising σ_{in} is the *stability* objective. Settings: population 30, 20 generations, random valid initialisation ($p = 0.5$ plus repair), two-point crossover, bit-flip mutation with rate $1/19$, duplicate elimination, and seeds 0, 1, 2.

Final choice ("within 1 SD" rule). Let b^* be the Pareto entry with the highest μ_{in}. The chosen chromosome is the cheapest entry whose accuracy lies within one standard deviation of the best:

$$\hat{b} = \arg\min_{b \in P}\ c(b) \quad \text{s.t.} \quad \mu_{in}(b)\ \geq\ \mu_{in}(b^*) - \sigma_{in}(b^*). \tag{4}$$

$\hat{b}$ is then refitted on the whole outer training set and scored once on the outer test fold.

Algorithm 1 Nested evaluation of the stability-aware GA for one outer fold

Require: outer training blocks $\mathcal{T}$, outer test block $\mathcal{V}$, seed s
$P \leftarrow$ 30 random valid chromosomes
for $g = 1$ **to** 20 **do**
  for each new chromosome b **do**
    $(\mu_{in}, \sigma_{in}) \leftarrow$ 3-fold CV of FBCSP+sLDA on $\mathcal{T}$ restricted to b
    objectives $\leftarrow (-\mu_{in},\ \sigma_{in},\ c(b))$
  end for
  $P \leftarrow$ NSGA-II survival (non-dominated sorting, crowding distance) of parents $\cup$ offspring
end for
$\hat{b} \leftarrow$ cheapest Pareto entry with $\mu_{in} \geq \mu_{in}(b^*) - \sigma_{in}(b^*)$
fit FBCSP+sLDA on all of $\mathcal{T}$ with $\hat{b}$; **return** balanced accuracy on $\mathcal{V}$

4.4 Baselines and comparison

- **FBCSP-all**: the same classifier with all 19 genes on (168 features, cost 1).
- **GA single-obj.**: the same GA optimising μ_{in} only; it picks the best inner accuracy.
- **EEGNet** [11]: a compact convolutional network (F1 = 8, D = 2, dropout 0.5, Adam, learning rate 10^{-3}, batch 32, up to 300 epochs with early stopping on a 20 % validation split of the training fold) on the 1–40 Hz epoch, with all 32 channels or the 8 candidate channels.

5 Evaluation Protocol

5.1 Data splits

- **Within session (block protocol):** leave-one-block-out, 4 folds per recording. Test trials are never temporal neighbours of training trials.
- **Cross-session:** train on session 1, test on session 2 (20 subjects).
- **Nesting:** the GA sees only the training blocks; the test fold is used exactly once per fold.

5.2 Metrics

Balanced accuracy (BA). For $K = 3$ classes with N_k test trials and TP_k correct predictions in class k,

$$\mathrm{BA} = \frac{1}{K}\sum_{k=1}^{K}\frac{TP_k}{N_k}, \qquad \text{chance} = 1/3. \tag{5}$$

BA is insensitive to the small class imbalance left after artefact rejection.

Unit of analysis. One score per subject: the three seeds are averaged first, then the sessions ($n = 22$ for block, $n = 20$ for cross-session). SDs are sample SDs over subjects.

Selection stability. The Kuncheva consistency index [13] between two selected subsets of size k from n features with overlap r is

$$\mathrm{KI} = \frac{rn - k^2}{k(n-k)}, \tag{6}$$

where 1 means identical subsets and 0 is the value expected for random subsets.

5.3 Statistical tests and pre-registration

The analysis plan was committed and pushed before the group results were analysed:

- **Primary hypothesis:** GA-NSGA > FBCSP-all, block protocol, one-sided Wilcoxon signed-rank test [14] over subjects, $\alpha = 0.05$.
- **Secondary family** (two-sided Wilcoxon, Holm-corrected [15]): S1 GA-NSGA vs GA single-obj.; S2 GA-NSGA vs FBCSP-all (cross-session); S3 GA-NSGA vs EEGNet-32; S4 GA-NSGA vs EEGNet-8.
- **Effect size:** matched-pairs rank-biserial correlation $r_{rb} = (W^+ - W^-)/(W^+ + W^-)$.
- **Per recording:** label-permutation test (1000 permutations), Holm-corrected over the 42 recordings.

With $n = 22$ and the observed SD of paired differences (0.045), the minimum detectable difference is about 0.024 BA (80 % power, one-sided $\alpha = 0.05$). All later analyses (Section 7) are labelled *exploratory*.

6 Results

6.1 Pre-registered primary and secondary tests

Table 2: Pre-registered tests (Wilcoxon signed-rank over subjects; Δ = first − second method).

Test	Comparison	n	Mean Δ	p	p_{Holm}	r_{rb}
Primary	GA-NSGA > FBCSP-all (one-sided), block	22	−0.007	0.788	–	−0.19
S1	GA-NSGA vs GA single-obj., block	22	−0.008	0.256	1.000	−0.28
S2	GA-NSGA vs FBCSP-all, cross-session	20	+0.003	0.616	1.000	+0.13
S3	GA-NSGA vs EEGNet (32 ch), block	22	+0.004	0.975	1.000	−0.01
S4	GA-NSGA vs EEGNet (8 ch), block	22	−0.002	1.000	1.000	−0.00
Sensitivity	S2 without subject 01	19	+0.006	0.463	–	+0.20

The primary hypothesis is not supported, and no secondary comparison is significant. The exploratory 95 % t-interval of the primary difference (−0.027 to +0.013) excludes gains above about 0.013 but cannot exclude losses up to about 0.03.

6.2 Accuracy, cost and generalisation

Table 3: Main results (subject level; mean ± sample SD). "> chance": recordings with permutation $p < 0.05$ after Holm correction. Drop = session-2 block BA − cross-session BA.

Method	Block BA (n = 22)	Cross-session BA (n = 20)	Drop	> chance	Cost / features
FBCSP-all	0.353 ± 0.044	0.337 ± 0.039	+0.027	2/42	1.00 / 168
GA-NSGA (proposed)	0.345 ± 0.025	0.340 ± 0.027	+0.002	2/42	**0.30 / 13**
GA single-obj.	0.353 ± 0.029	0.328 ± 0.027	+0.028	3/42	0.48 / 39
EEGNet (32 ch)	0.341 ± 0.028	0.325 ± 0.029	+0.013	2/42	– / 3107 params
EEGNet (8 ch)	0.347 ± 0.031	0.334 ± 0.033	+0.004	1/42	– / 2723 params

All five methods are within 0.012 BA of each other on the block protocol, and none is more than 0.02 above chance (Figure 1). Only 1–3 of 42 recordings per method are significant after Holm correction. GA-NSGA matches the baseline with about 8 % of its features and has the smallest cross-session drop (Figure 2), but its cross-session accuracy is not higher (S2).

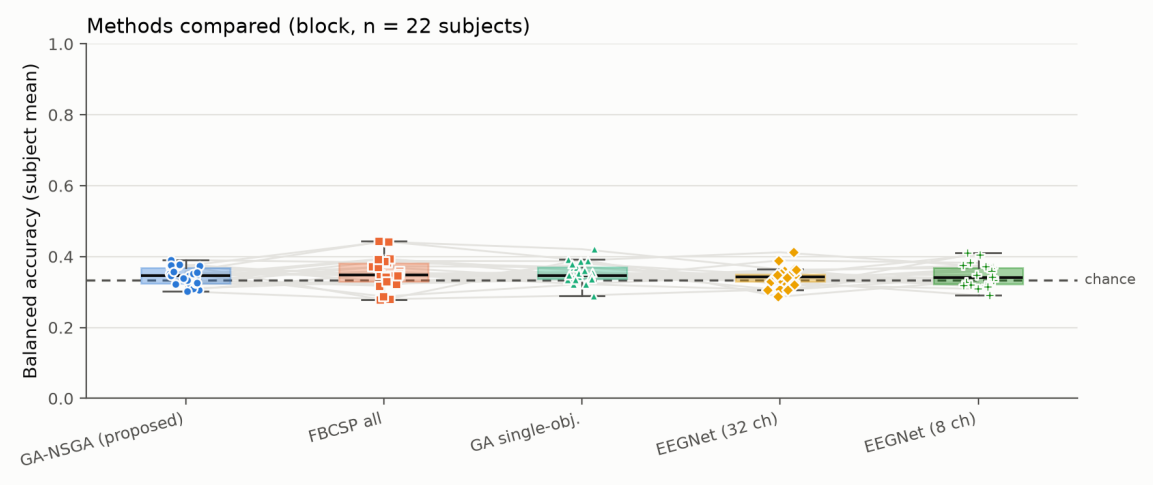


Figure 1: Subject-level balanced accuracy, block protocol (n = 22). One dot per subject; the dashed line is chance. The pre-registered primary test is not significant.

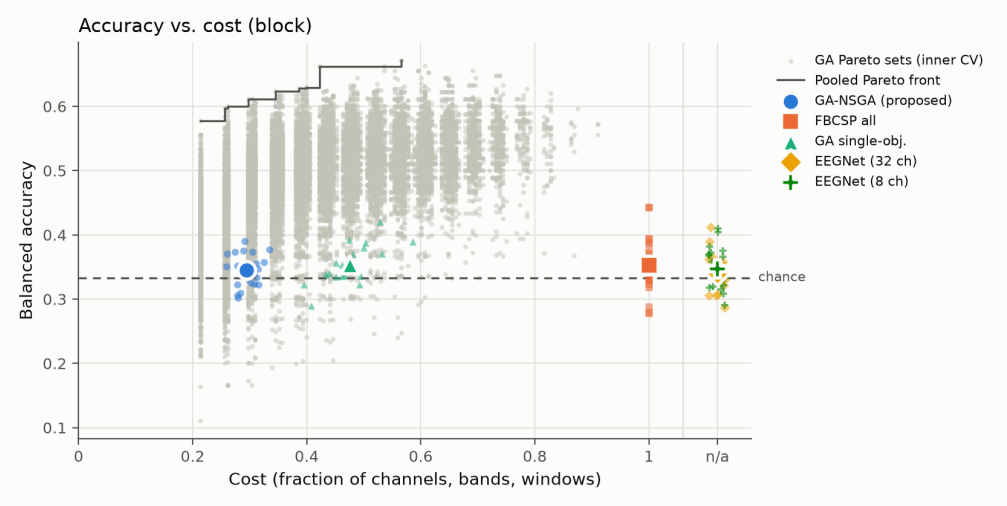


Figure 2: Accuracy against cost (block protocol). Grey points are the GA's inner-CV Pareto entries, which are optimistic; coloured points are outer-test subject means. GA-NSGA reaches FBCSP-all's accuracy at about 30 % of the cost.

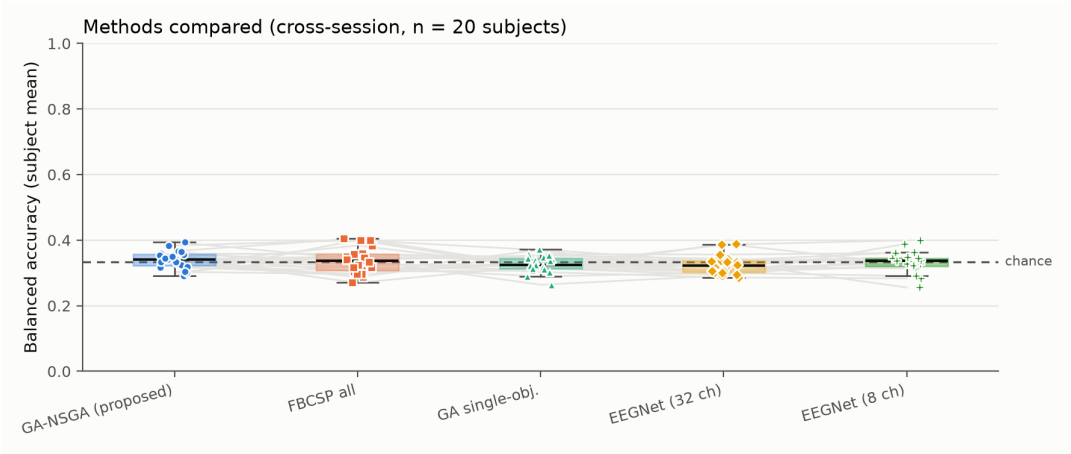


Figure 3: Cross-session balanced accuracy (train on session 1, test on session 2; $n = 20$).

6.3 What the GA selects

GA-NSGA selects on average 2.7 channels, 1.3 bands and 1.6 windows. Selection is nearly flat across genes (18–43 %; most frequent Pz 43 %, Oz 38 %, P7 35 %; Figure 4). Selection stability is close to chance and no higher than for the accuracy-only GA (Table 4).

Table 4: Selection and stability of the chosen chromosomes (block protocol; Kuncheva index, mean over recordings).

Method	Ch / bands / windows	KI (all pairs)	KI (same fold, other seed)	KI (same seed, other fold)
GA-NSGA	2.7 / 1.3 / 1.6	0.078	0.163	0.072
GA single-obj.	3.8 / 2.0 / 3.2	0.084	0.254	0.058

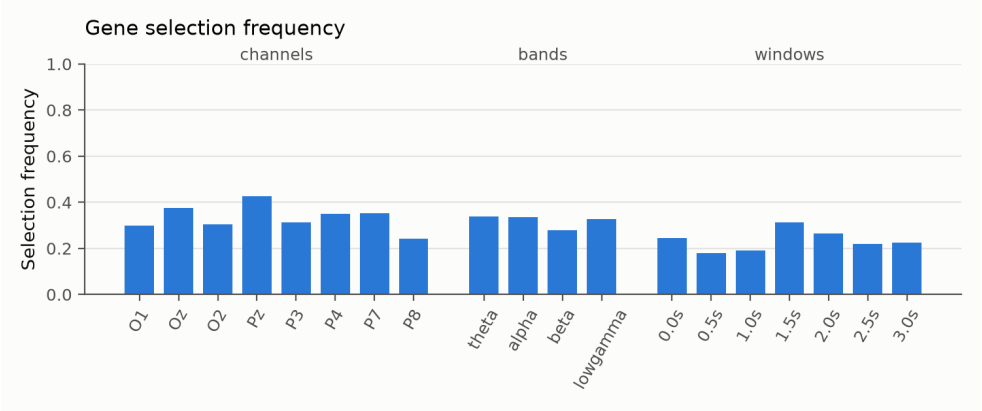


Figure 4: How often GA-NSGA selects each gene (504 chosen chromosomes: 126 block runs × 4 folds). The near-flat profile shows no consistent preference.

7 Exploratory Analyses

The following checks were *not* part of the analysis plan. They test why the GA does not help.

7.1 Random chromosomes at matched cost

For every outer fold, 50 random chromosomes with exactly the same numbers of channels, bands and windows as the GA's choice were scored on the same test fold with the same classifier.

- GA-NSGA vs random: +0.004 (p = 0.42); GA single-obj. vs random: +0.007 (p = 0.20).
- The GA's mean percentile among the random draws is 0.51 / 0.52 (0.5 = a typical random subset).
- Random subsets at 30 % of the cost lose only 0.011 against FBCSP-all (n.s.).

The GA search adds nothing beyond the cost reduction (Figure 5).

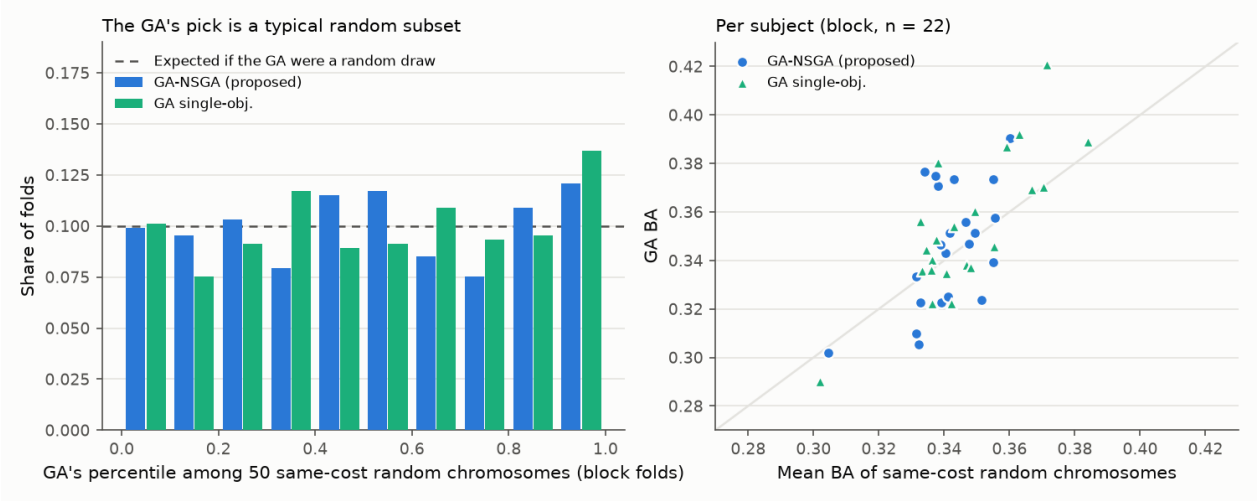


Figure 5: Left: the GA's percentile among 50 same-cost random chromosomes per fold; the flat histogram means its pick is a typical random subset. Right: per subject, GA accuracy vs the mean accuracy of the random subsets.

7.2 Alternative chooser rules

Seven rules were applied to the stored GA fronts: best inner accuracy, 0.5/1/2 SD, standard error, knee point and minimum cost. They give 0.338–0.351 on the block protocol. None differs from the contract rule or from FBCSP-all (all $p \geq 0.28$; Figure 6). These are test-set scores, so choosing a rule by them would be selection on the test set.

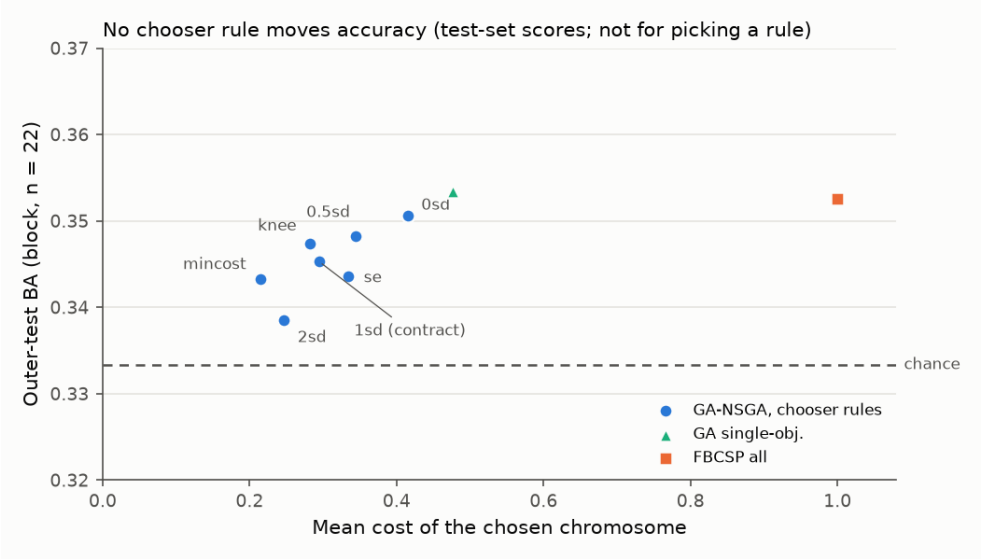


Figure 6: Outer-test accuracy against cost for alternative chooser rules on the stored GA-NSGA fronts.

7.3 Training versus test accuracy

Each fold's classifier was refitted on its outer training set and also scored on those training trials (Table 5, Figure 7). EEGNet trained for 300 epochs without early stopping reaches a training accuracy of 1.0 while its validation accuracy stays at chance (Figure 8).

Table 5: Training, inner-CV and test balanced accuracy (block protocol). For EEGNet the middle column is the validation accuracy.

Method	Training BA	Inner-CV / validation BA	Test BA
FBCSP-all	0.999	0.342	0.353
GA single-obj.	0.751	0.568	0.353
GA-NSGA	0.576	0.512	0.345
EEGNet (32 ch, P4 recipe)	0.639	0.364	0.350
EEGNet (32 ch, 300 epochs, no early stopping)	1.000	0.335	0.327

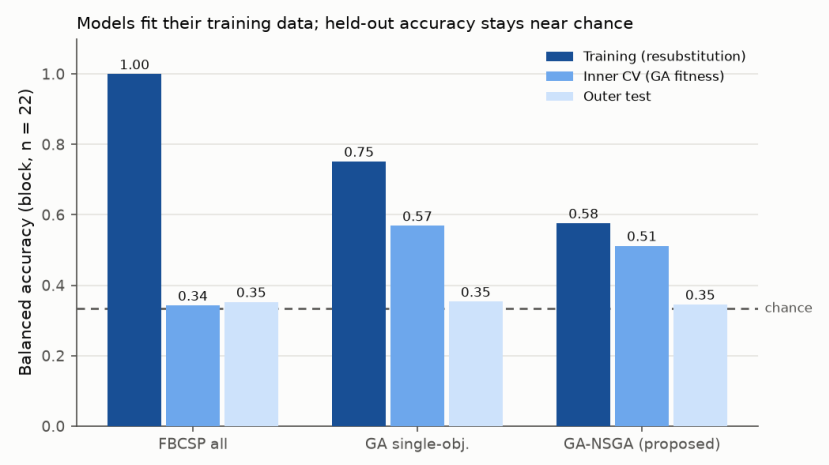


Figure 7: Training vs inner-CV vs test balanced accuracy (block protocol). Every model fits its training trials but generalises to about chance +0.02; the GA's inner-CV score is inflated by selection.

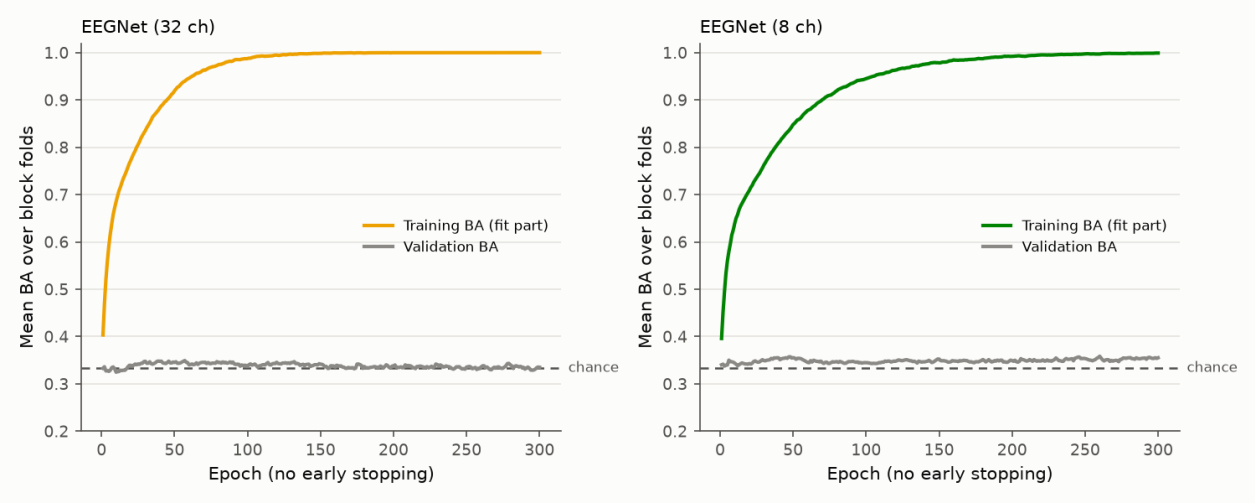


Figure 8: EEGNet learning curves without early stopping (300 epochs, block folds, seed 0): training accuracy reaches 1.0 while validation accuracy stays at chance.

7.4 Trial-position confound

A classifier that sees only the trial's position within its block, with no EEG, predicts the class across sessions at 0.397 ($p < 0.001$; 0.390 without subject 01). The reason is that 30-trial block orders recur across sessions for 10 of 20 subjects. The EEG methods do not follow this signal: they show no gain on the shared-block subjects (FBCSP-all 0.337 vs 0.337) and trial-level agreement $\kappa \approx 0$ (Figure 9). For subject 01, position and class are perfectly confounded, so its cross-session result cannot be interpreted.

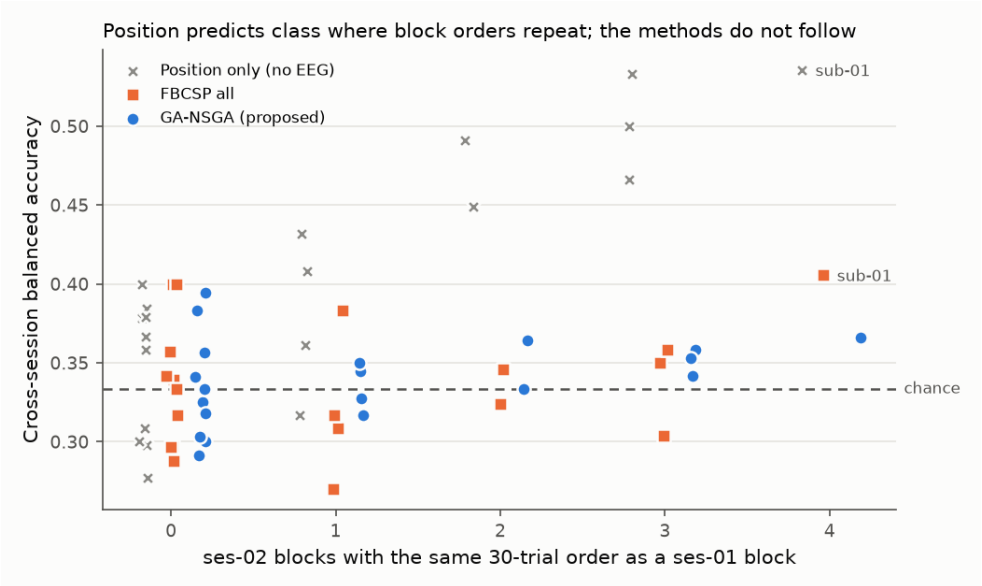


Figure 9: Position-only control (cross-session) against the number of session-2 blocks whose order also occurs in session 1.

7.5 Random-split inflation

Random stratified 5-fold splits mix temporally adjacent trials into training and test, which can let slow drifts look predictive. Here they do not inflate any method: the change from block to random splits is −0.002 to −0.011 (all n.s.; Figure 10).

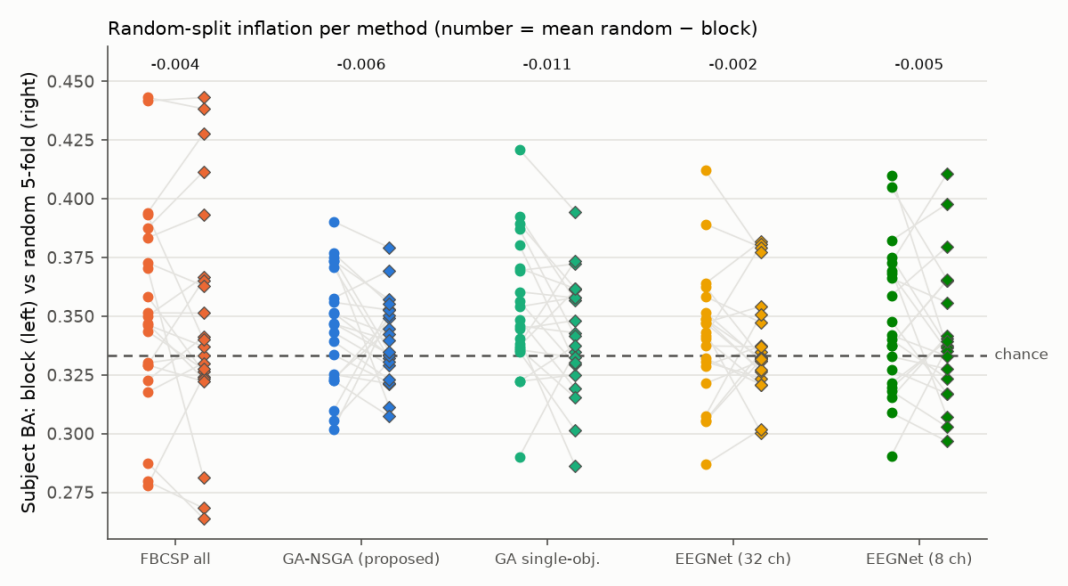


Figure 10: Subject accuracy under leave-one-block-out (left) and random 5-fold splits (right).

7.6 Perception (viewing) period

On the 3–7 s viewing period, where the participant sees the shape, all methods decode well above the imagery period (Table 6, Figure 11). Here GA-NSGA is significantly worse than FBCSP-all (−0.025, $p = 0.004$) and than GA single-obj. (−0.018, $p = 0.04$), and the chooser rule matters (0.377–0.421 across rules).

Table 6: Viewing period vs imagery (block protocol, $n = 22$).

Method	Imagery BA	Viewing BA	Difference	p	Recordings > chance (Holm)
FBCSP-all	0.353	0.435 ± 0.069	+0.083	< 0.001	12/42
GA single-obj.	0.353	0.428 ± 0.074	+0.074	< 0.001	20/42
GA-NSGA	0.345	0.410 ± 0.048	+0.065	< 0.001	15/42

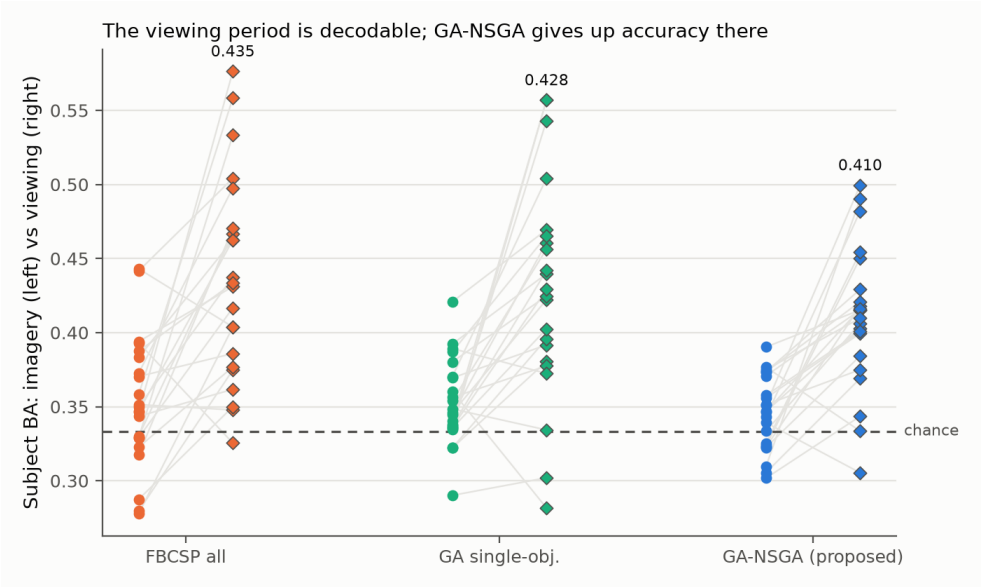


Figure 11: Imagery (left) vs viewing (right) per subject. With a real signal, the GA's cost pressure costs accuracy.

8 Discussion

The overfitting gap. Every model fits its training trials, from 0.58 training accuracy (GA-NSGA, 13 features) to 1.00 (FBCSP-all with 168 features, and EEGNet without early stopping), yet all land at about 0.35 on unseen trials. Shrinking the model to 13 features lowers the training fit but not the test accuracy. The limiting factor is the information in the imagery EEG, not the classifier.

Why the GA cannot help. The GA's fitness is a 3-fold inner CV on about 90 trials, which is dominated by noise [16]. Without selection this estimate is honest: FBCSP-all's inner CV is 0.342 against a test accuracy of 0.353. The GA takes the best of about 600 such estimates per fold and therefore selects lucky chromosomes. This is the *winner's curse* of model selection [17], and it inflates the inner accuracy by +0.17 (GA-NSGA) and +0.21 (GA single-obj.). The stability objective σ_{in} comes from the same three folds and is equally noisy, so it does not make the selection reproducible (KI 0.08). Selection can only help when some components carry more class information than others *and* the inner CV can detect it; on the imagery data neither holds.

When there is a signal. On the viewing period the expected trade-off appears: smaller models lose accuracy, and the GA's cost pressure costs 0.025 BA for a 67 % cost reduction. The stability-aware GA is therefore a *cost-reduction* tool with a measurable price, not a way to recover a weak signal.

Comparison with the literature. Gao et al. [6] report 75.1 % for EEGNet on these three figure classes with a within-subject protocol. Under our pipeline EEGNet reaches 0.341 (block) and 0.339 with random 5-fold splits, which the dataset notes identify as the authors' protocol, so split leakage does not explain the difference. Differences in epoching, preprocessing or training details remain possible. We could not reproduce the published figure, and reconciling the two pipelines is future work. The literature values in Table 1 come from different datasets, classes and protocols and are not directly comparable.

9 Limitations

- **Class set:** three classes (circle, square, pentagram) instead of the proposal's four, because no public dataset contains triangle and hexagon in this paradigm.
- **Weak signal and sample size:** group accuracy is at most 0.02 above chance; with $n = 22$ only differences of about 0.024 BA or more are detectable.
- **Fixed design:** the candidate channels, bands, windows, GA budget (30×20) and EEGNet recipe were frozen before the results; other search spaces were not tried.
- **Small inner CV:** three folds of about 30 trials are the GA's only fitness signal.
- **Trial-order confounds** in the dataset (Section 7.4) affect any cross-session or cross-subject analysis.
- **Artefact handling** uses the 8 candidate channels only (no EOG or ICA), so 32-channel EEGNet sees frontal blink activity.
- **Exploratory analyses** were not corrected for multiplicity across checks.

10 Conclusion and Future Work

On 22 subjects, a stability-aware NSGA-II that jointly selects EEG channels, frequency bands and time windows **did not outperform** filter-bank CSP on all components for imagined-shape decoding (0.345 vs 0.353 balanced accuracy, $p = 0.79$). It also did not outperform an accuracy-only GA or EEGNet. It delivers the same near-chance accuracy at about 30 % of the cost, but its selections are no better than random subsets of the same size and are not reproducible. Every model memorises its training data and generalises to about chance +0.02, so the imagery signal of this dataset, not the classifier, is the bottleneck. When a real signal exists (the perception period), the GA's cost reduction has a significant accuracy price.

Future work.

- Evaluate the method on a dataset with the proposal's four shapes (circle, square, triangle, hexagon) and more trials per class.
- Reconcile our EEGNet pipeline with the 75.1 % reported by Gao et al.
- Use fitness estimates that are robust to the winner's curse (repeated inner CV, nested stability selection) or multi-session training.
- Apply the selection where a signal is strong (perception, or imagery with feedback training) and the cost of channels matters, e.g. low-channel wearable BCIs.

A Reproducibility and Software

- **Code:** Python package with fixed module contracts (`src/`: io, preprocess, cv, fbcsp, ga, eegnet, analysis), 200 automated tests, and release tag `v1.0`.
- **Determinism:** GA results are bit-identical across the three team machines, and EEGNet is deterministic on CUDA. Every result file records its code commit, configuration hash (`08e124167b`) and host.
- **Pre-registration and deviations:** the analysis plan was pushed before the group results were analysed. One machine had run the grid locally before the plan was pushed; those files were never used, and the official runs are bit-identical to them. All deviations are documented.
- **Run times** (i5-13450HX, RTX 3050 6 GB): cache build ≈7 min, baseline 9 s, full GA grid 5.3 min, EEGNet 49 min.

B Team Contributions

Member	Responsibilities
Abhay Nagesh Rao (26CSM1R01)	Cross-validation splits, FBCSP + shrinkage LDA, statistical analysis and figures; GA runs for subjects 09–16.
Pandya Aditya Deepakkumar (26CSM1R02)	Data download, BIDS reading, preprocessing and caches, integration, FBCSP baseline, pre-registered analysis plan; GA runs for subjects 01–08.
Mulakalapalli Jayachandra Siddhartha (26CSM1R07)	Genetic algorithms (NSGA-II, single-objective), EEGNet, GA runs for subjects 17–22 and all EEGNet runs, exploratory analyses, final results and release.

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